

Operator-based approaches to the Riemann Hypothesis with a focus on construction and testing.

Executive Summary

The **Riemann Hypothesis (HR)** states that every non-trivial zero of the Riemann zeta function has a real part $1/2$. The **Hilbert–Pólya approach** attempts to prove HR by constructing a **self-adjoint operator**. H whose spectrum is (or determines) the heights γ of the zeros $1/2 + i\gamma$: self-adjointness $\Rightarrow$ real spectrum $\Rightarrow$ zeros on the critical line. This strategy is motivated by rigorous analogies already solved (Selberg case) and by statistical evidence (Montgomery–Odlyzko/GUE). [1]

What is already established: (i) the formalism ζ , functional equation and ξ zero counting (Riemann–von Mangoldt) form the “contract” that any candidate operator must respect; (ii) there are models where a $N(T)$ **self-adjoint Laplacian** forces an “HR type” on geometric zetas (Selberg); (iii) there are influential proposals for ζ HR itself, especially the [2]entity[2] [“people”, “Alain Connes”, “mathematician”] [2] program (non-commutative geometry and trace formula) and the [2]entity[2] [“people”, “Michael Berry”, “physicist”] [2]–[2]entity[2] [“people”, “Jonathan P. Keating”, “mathematician”] [2] program ($H = xp$ semiclassical Hamiltonian), but none close HR. [2]

The “Connes bottleneck” today is quite specific: Connes and [2]entity[2] [“people”, “Caterina Consani”, “mathematician”] [2] show a **conceptual reason for the positivity of the Weil functional** in the **single Archimedean case**, expressing the difference between the Weil distribution and a **compressed trace** (“Sonin trace”) via prolate spheroidal functions and controlling it with Hermitian Toeplitz matrices; however, the complete extension to the **semi-local/global case** (where positivity would imply HR) is not yet complete. [3]

This report delivers: (a) explicit mathematical requirements for a candidate operator; (b) a technical survey of the main proposals (Connes, xp Berry–Keating, quantum graphs, integral kernels, de Branges, Deninger); (c) reproducible computational protocols for testing “toy operators” against the [2]entity[2] [“people”, “Andrew Odlyzko”, “mathematician”] [2] data; (d) a clear formulation of what would need to be proven to “close” the Connes program gap — without claiming proof (the problem remains open). [4]

Formal statement and definitions

For $\Re(s) > 1$, zeta is

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s},$$

and satisfies **Euler's product**

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}} \quad (\Re(s) > 1),$$

which encodes primes. [5]

The analytical continuation (except for the simple pole in $s = 1$) and the functional equation become cleaner via the completed function.

$$\xi(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s), \quad \xi(s) = \xi(1-s),$$

which is whole. [6]

The **trivial zeros** of ζ are in $s = -2, -4, -6, \dots$. The **non-trivial zeros** are in the critical strip $0 < \Re(s) < 1$. The **Riemann Hypothesis** (standard formulation of the [identity](#)["organization", "Clay Mathematics Institute", "millennium prize org"]^[7]) is:

$$\zeta(\rho) = 0, \quad 0 < \Re(\rho) < 1 \quad \Rightarrow \quad \Re(\rho) = \frac{1}{2}.$$

[7]

Define $\mathcal{E}(t) = \xi(1/2 + it)$. It is real and even; the non-trivial zeros correspond to the real zeros $\pm \gamma_n$ of $\mathcal{E}$ if HR is true. The zero count up to height T is (principal Riemann–von Mangoldt terms):

$$N(T) = \#\{\rho = \beta + i\gamma: 0 < \gamma \leq T\} \approx \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}.$$

This law is the reference for “Weyl law” for any candidate operator. [8]

Historical context and milestones with a spectral focus

The original text by [identity](#)["people", "Bernhard Riemann", "mathematician"]^[9] (1859, translated by [identity](#)["people", "David R. Wilkins", "translator"]^[10]) already points out that zeros of a complex function control the distribution of primes. [9]

The Hilbert–Pólya intuition (frequently associated with [identity](#)["people", "David Hilbert", "mathematician"]^[11] and [identity](#)["people", "George Pólya", "mathematician"]^[12]) gains concrete basis in the “Selberg universe”: geometric trace and zeta formulas where the spectrum of a self-adjoint Laplacian forces “RH-type” alignment. This is the model case where “a certain operator” can impose a critical line. [10]

Spectral statistical evidence comes in with [identity](#)["people", "Hugh L. Montgomery", "mathematician"]^[13] (pair correlation) and Odlyzko (spacings), connecting zeros to GUE statistics and the imaginary of “chaotic quantum spectra”. [11]

In the modern era, Connes' program reinterprets the explicit formula as a trace formula within non-commutative geometry in the space of classes of adeles, viewing critical zeros

as "absorption spectrum" and zeros outside the line as resonances; Berry–Keating, in turn, proposes it $H = xp$ as semiclassical dynamics that correctly reproduces the mean term of $N(T)$. [12]

Recent relevant results include: (i) [Terence Tao, "mathematician"] and [Brad Rodgers, "mathematician"] proving $\Lambda \geq 0$ (de Bruijn–Newman constant), reinforcing the narrative of “HR, if true, is tight”; (ii) rigorous verification of HR even $3 \cdot 10^{12}$ by [Dave Platt, "mathematician"] and [Tim Trudgian, "mathematician"]; (iii) Connes–Consani (2020/2021) establishing positivity in the Archimedean case and describing the bridge to the semi-local case; (iv) preprints (2024–2025) proposing “Hilbert–Pólya” Hamiltonians with boundary conditions involving ζ , still without broad consensus/validation. [13]

Timeline

Title Milestones in Operator-Based Approaches

1859: Riemann links $\zeta(s)$ to zeros and primes

1956: Selberg and the trace formula (solved spectral model for geometric zetas)

1973: Montgomery: pairwise correlation (under RH)

1987: Odlyzko: zero spacings $\sim$ GUE

1999: Connes: non-commutative trace and absorption spectrum

1999: Berry–Keating: $H=xp$ dynamics and $N(T)$ mean term

2003: Voros: spectral functions constructed on zeros (Selberg/Weil parallel)

2009–2011: formal obstructions to direct quantizations of the BK operator in simple spaces

2014: quantum graphs mimic the oscillatory part of the zero density

2018: Rodgers–Tao: constant of de Bruijn–Newman ≥ 0

2021: Connes–Consani: positivity (Archimedean locus) via compressed trace and Toeplitz

2021: Platt–Trudgian: rigorous verification of zeros up to $3 \cdot 10^{12}$

2026: Connes: survey with variational method and new approximations on the critical line

[14]

Mathematical requirements for a candidate operator

A serious Hilbert–Pólya candidate must simultaneously satisfy:

Self-adjointness and explicit domain. There must exist a Hilbert space $\mathcal{H}$ and a linear operator $H: \mathcal{D}(H) \rightarrow \mathcal{H}$ with a dense domain $\mathcal{D}(H)$ such that H is self-adjoint (or essentially self-adjoint). Without this, complex eigenvalues (and therefore zeros outside the critical line) are not excluded. This is the “stage II” that modern proposals recognize as the bottleneck. [15]

Weyl's law is compatible with Riemann–von Mangoldt. If the goal is that $\text{spec}(H) = \{\gamma_n\}$ (or a controlled subset), then the spectral count must obey

$$N_H(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + \frac{7}{8},$$

at least in the main term (the middle term). Berry–Keating reproduces this term by semiclassical arguments, but this is not enough: a well-defined operator is needed whose counting and corrections are correct. [16]

Trace formula equivalent to the explicit formula. The operator must produce (possibly in a regularized way) an identity that connects the spectral side (zeros/eigenvalues) to the “geometric/dynamic” side (primes such as periods $\log p$ and repetitions $m \log p$), analogously to Selberg's trace formula. Connes' program claims to rewrite the explicit formula as a trace formula (semi-local/global) for the compressed scale action. [17]

Regularized trace and regular determinants. In general, e^{itH} or resolvents are not a trace class; a regularized trace/distribution formalism consistent with the explicit formula is required. This appears explicitly in Connes (distributions and “Sonin trace”) and in approaches to regularized determinants/cohomology (lines à la Deninger) and spectral functions over zeros (Voros). [18]

Concrete proposals and constructions with critical analysis.

Below is a technical overview (showing “how concrete” and “where it breaks down”), focusing on the building blocks that allow for construction and testing.

Connes–Consani structure: compressed scale action and Weil positivity

Connes (1999) formulates a spectral interpretation in non-commutative geometry in the space of classes of adeles; Connes–Consani (2020/2021) attack the point of **positivity of the Weil functional** in the Archimedean case: the positivity comes from the trace of the compressed scaling action in the orthogonal complement of the range of cutoff projections; the difference between the Weil distribution and this trace (“Sonin trace”) is expressed by prolate spheroidal functions and controlled using Hermitian Toeplitz matrices. [3]

Strength: it is the most “structural” path where the explicit formula appears as a feature, and there is a concrete mechanism (compression + Toeplitz) that demonstrates positivity in a case. [19]

Weakness/gap: it lacks establishing the complete analogue in the semi-local/global case; the article itself states that the tools make sense overall and that “Weil positivity implies RH”, but it does not close the global step. [20]

Berry–Keating $H = xp$ and formal obstructions

The Hamiltonian $H = xp$ (with regularizations) semiclassically reproduces the mean term of the zero count; however, when quantized on simple spaces, the candidate BK has a continuous spectrum and cannot produce the discrete eigenvalues of the zeros; even on compact graphs, the resulting Weyl law does not fit the Riemann–von Mangoldt law. This

is explicitly proven by [entity["people", "Sebastian Endres", "mathematical physicist"]] and [entity["people", "Frank Steiner", "mathematical physicist"]]. [21]

Strength: explains the middle term $N(T)$ with a simple model; connects to the intuition of hyperbolic chaos. [22]

Weakness: does not deliver a self-adjoint operator with the correct spectrum; the “cutoffs/BC adjustment” becomes non-canonical and tends to reintroduce circularity. [23]

Quantum graphs and Laplacian graphs: testable proposals

There are two useful subprograms:

- **Mimicking the oscillatory part via lengths $\log p$.** Kuipers et al. construct infinite families of graphs (“butterfly graphs”) whose oscillatory term of the density of states coincides with the sum over primes and repetitions $m \log p$. The text itself points out that the average part of typical graphs does not have the correct logarithmic dependence and that the global spectrum becomes dense/divergent without additional controls. [24]
- **Obtaining “non-standard” Weyl laws with logarithmic component.** Egger–Steiner study an infinite graph with lengths $l_n = \pi/n$, obtaining spectral count $\mathcal{N}(\lambda) \sim \frac{\sqrt{\lambda}}{2} \ln \lambda + O(\sqrt{\lambda})$, and derive exact trace and “Selberg-like” zeta formulas with equation functional and true “RH analog”. This is important as proof-of-concept: graphs can produce counts with log. [25]

Strength: They are concrete models with self-adjoint Laplacians and computable traces; excellent for experiments and for “reverse engineering” of the mean/oscillatory term. [26]

Weakness: There is still no single canonical graph/operator whose spectrum is exactly $\{\gamma_n\}$ and whose trace formula coincides with the complete explicit formula without “leftover”. [27]

Integral and determinant nuclei: Voros and Deninger

[entity["people", "A Voros", "mathematical physicist"]] systematizes “zeta functions over Riemann zeros”, making explicit parallels with spectral zetas (Selberg/Minakshisundaram–Pleijel) and with Weil’s explicit formula; this formalizes “how a spectral determinant associated with zeros should look” and discusses regularizations. [28]

The “cohomological/regularized determinant” line associated with [entity["people", "Christopher Deninger", "mathematician"]] (here via expository notes by Matthias Flach) seeks a Frobenius-type operator whose regularized determinant reproduces zetas, inspired by the Weil/Deligne success in function fields; but remains programmatic. [29]

Branges and Hilbert spaces of entire functions

[entity["people", "Louis de Branges", "mathematician"]] proposes framing HR within the framework of Hilbert spaces of entire functions $H(E)$, where invariance/positivity axioms

impose strong restrictions on zeros. It is a similar approach in "operational spirit" (space structure and natural operators), but closing the case ξ requires verifying conditions that have not yet been established and there is critical literature on difficulties of positivity in the desired case. [30]

Equivalent criteria connectable to the operator pathway.

- **Nyman–Beurling / Báez-Duarte** : HR is equivalent to a density statement in a Hilbert subspace L^2 , which speaks directly to positivity/compression programs (a “functional-analytic” form of HR). [31]
- **Li's criterion** (via Li's coefficients): HR is equivalent to the non-negativity of a sequence defined from zeros; although it is another "packaging" of the same problem, it generates numerical tests and quadratic forms (bridge with positivity). [32]
- **Weil/Deligne analogy** : in function fields, analogous HR is proved via Frobenius/cohomology (Weil conjectures), motivating the dream of a “hidden Frobenius” for $\mathbb{Q}$. [33]

Computational protocols and reproducible scripts

Data and tools

Recommended dataset: “first 100,000 zeros” in Odlyzko tables (file zeros1), which is the standard basis for gap statistics and numerical tests. [34]

Tools:

- Python + NumPy/SciPy/Matplotlib (statistics and simulations),
- mpmath (high-precision calculation)
- `Entity["organization","SageMath","math software"]` (includes packages of zeros and L-functions),
- PARI/GP (sums/Dirichlet series),
- Arb/interval arithmetic (for rigorous checks in the Platt–Trudgian style). [35]

Computational resources: **not specified** . Realistic estimates:

- Laptop: gaps/CDF/pairwise correlation with 100k–2M zeros (minutes–hours).
- Workstation: scan dozens of toy operators and simulate large random matrices (hours–days).
- Cluster: heavy experiments with regularized traces and interval checks (days–weeks). [35]

Requested charts

Below are reproducible graphs from Odlyzko's `zeros1` file . They serve as a "comparison standard" for toy operator spectra: simply replace them γ_n with the eigenvalues λ_n of your operator and repeat the routines.

Histogram of normalized gaps vs. GUE (Wigner Surmise)

Histogram

Paired

correlation (local estimate) vs. GUE curve. $1 - \left(\frac{\sin(\pi u)}{\pi u} \right)^2$

counting $N(T)$ vs. Riemann–von Mangoldt principal term

$N(T)$ vs. RvM

(to see the deviation)

Error $N(T) - N_{\text{RvM}}(T)$

Downloads:

- [gap_hist.png](#)
- [pair_correlation.png](#)
- [N_vs_RvM.png](#)
- [N_error.png](#)

Data source: Odlyzko zero tables (`zeros1`). [36]

Step-by-step experimental protocol

Protocol A — benchmark (real zeros).

- 1) Download `zeros1` (Odlyzko).
- 2) Calculate normalized gaps s_n and compare with Wigner-GUE (hist, CDF, KS).
- 3) Calculate a pairwise correlation estimate (window) and compare with $1 - \left(\frac{\sin(\pi u)}{\pi u} \right)^2$.
- 4) Calculate $N(T)$ and compare with the Riemann–von Mangoldt principal term.

This reproduces the graphs above and provides a baseline for evaluating toy operators. [37]

Protocol B — Toy Operator (Replace Eigenvalues).

- 1) Choose a candidate operator H and compute a set of eigenvalues $\lambda_1, \dots, \lambda_M$.
- 2) Replace it $\gamma_n \leftarrow \lambda_n$ in the Protocol A code.
- 3) Compare: (i) average term count $N_H(E)$; (ii) gap statistics; (iii) pairwise correlation.

4) If the toy operator fails the average term, adjust that first (it's a "prerequisite"); if it passes the average term but fails the gaps, the problem is in the oscillatory/trace part.

Protocol C — test trace identity (only models where the trace formula is known).

For quantum graphs (Laplacian on graph) and for Egger–Steiner models, there are explicit trace formulas; you can numerically compare the “periodic” side (sum over orbits) with the spectral side (sum over eigenvalues, with test function). This is the computational analogue of the explicit formula. [38]

Script skeleton (replaceable)

```
import numpy as np, math
import scipy.special
import matplotlib.pyplot as plt

# === 1) Input: data ===
# Option 1: Odlyzko zeros
gam = np.loadtxt("zeros1.txt") # 100000 values

# Option 2: replace with eigenvalues of your toy operator
# gam = np.sort(lambdas) # lambdas = eigenvalues(H)

# === 2) Normalized gaps ===
gaps = gam[1:] - gam[:-1]
scale = np.log(gam[:-1] / (2*math.pi)) / (2*math.pi)
s = gaps * scale

# === 3) GUE (Wigner surmise) curves ===
def p_gue(x):
    return (32/math.pi**2) * x**2 * np.exp(-4*x**2/math.pi)

def F_gue(x):
    return scipy.special.erf(2*x/math.sqrt(math.pi)) - (4*x/math.pi)*np.exp(-4*x**2/math.pi)

# === 4) Plots ===
bins = np.linspace(0,3,61)
hist, edges = np.histogram(s, bins=bins, density=True)
centers = 0.5*(edges[:-1] + edges[1:])
x = np.linspace(0,3,1000)

plt.figure()
plt.bar(centers, hist, width=edges[1]-edges[0], alpha=0.7,
label="empirical")
plt.plot(x, p_gue(x), linewidth=2, label="GUE (Wigner)")
plt.legend(); plt.xlabel("s"); plt.ylabel("density"); plt.show()

# === 5) Count N(T) vs RvM (main term) ===
T = gam
```



```
Nemp = np.arange(1, len(T)+1)
NRvM = (T/(2*math.pi))*np.log(T/(2*math.pi)) - T/(2*math.pi) + 7/8
```

```
plt.figure()
plt.plot(T[::200], Nemp[::200], label="N(T) empirical")
plt.plot(T[::200], NRvM[::200], linewidth=2, label="RvM main term")
plt.legend(); plt.xlabel("T"); plt.ylabel("N(T)"); plt.show()
```

The “Connes problem” in one line

Connes–Consani show in the Archimedean locus that Weil positivity has a conceptual origin: it comes from the **compressed scale action trace** in the orthogonal complement of the cutoff; they control the difference between the Weil functional and this trace (Sonin trace) with prolate + Toeplitz functions. What is missing is extending this mechanism to the **semi-local/global framework**, where (as they claim) positivity would imply HR. [3]

What would be a “chain of slogans” that closes HR via Connes

One operational (conditional) way to close would be to prove the following items:

- **Global identification lemma:** for an explicit class of test functions f , the Weil functional $W(f)$ coincides with a regularized/compressed trace $\text{Tr}_{\text{reg}}(T_f)$ defined in the semi-local structure (scaling action + cutoff projections) of the Connes program. [17]
- **Global positivity lemma:** show $W(f) \geq 0$ for all f admissible (equivalent to “Weil positivity”). Connes–Consani prove a (Archimedean) case and indicate the tools for the semi-local case. [20]
- **Equivalence lemma:** rigorously justify that $W(f) \geq 0$ in the test class implies HR for ζ (or even for a family of L -functions) within the formalism (this equivalence is part of the “Weil positivity $\Rightarrow$ HR” framework mentioned by Connes–Consani). [17]
- **Error control/cutoffs lemma:** dealing with infrared/ultraviolet terms and boundary passage (cutoffs are a central part of Connes' modern formalism, including in the 2026 survey with variational method and approximations via truncated Euler products). [40]

If these lemmas are proven, the gap closes and HR follows within this program. This is the correct meaning of “solving by Connes”: it is not about finding more zeros, it is about proving the global principle of positivity/trace. [41]

Where can you “really work” (without promising proof)?

Three plausible and testable lines of attack:

1) **Generalize the Toeplitz/prolate control of the Archimedean to the semi-local.** This means constructing the semi-local analogue of cutoff projections and proving a matrix/operational inequality that replaces the Toeplitz step of the Archimedean. One can

start numerically (finite approximations) before attempting the infinite-dimensional argument. [42]

2) **Explore the “prolate operator” and UV/IR spectra as a concrete bridge.** Connes–Moscovici (and related texts) suggest a structural role for the prolate operator and associated self-adjoint extensions; this may give a more explicit model for the “operational” part of the program. [43]

3) **Use Connes’ (2026) variational method as a guided heuristic.** In the 2026 survey, Connes describes an extremization procedure of a quadratic form (linked to the Weil functional) that approximates zeros with high precision using few primes, and proves that the approximants lie on the critical line. This does not prove HR, but suggests that there is an exploitable convexity/positivity structure — a possible “laboratory” for conjecturing what should be proven in the infinite limit. [44]

Essential resources for anyone who wants to tackle this seriously.

- Riemann 1859 (Wilkins translation). [5]
- Bombieri (official problem description) and Clay's page. [45]
- Connes 1999 (non-commutative trace and zeros) + Connes–Consani 2020/2021 (Archimedean positivity). [46]
- Montgomery and Odlyzko (spectral statistics of zeros). [11]
- Endres–Steiner (obstructions for x_p simple spaces; graphs). [47]
- Kuipers et al. (graphs mimicking the oscillatory part). [48]
- Equivalent criteria: Báez-Duarte/Nyman–Beurling; Conrey (includes Li/Weil criterion). [49]

Honest conclusion: today, “closing once and for all” the Connes loophole would be equivalent to a new HR proof. What can be done with rigor and immediate utility is: (i) formalize the chain of lemmas above; (ii) implement numerical experiments that test finite versions (Toeplitz/prolate/compressed trace); (iii) look for a semi-local generalization of the positivity control that already works in the Archimedean case. [41]

[1] [6] [7] [8] [45] Problems of the Millennium: the Riemann Hypothesis

https://www.claymath.org/wp-content/uploads/2022/05/riemann.pdf?utm_source=chatgpt.com

[2] [12] [16] [46] Weil positivity and Trace formula the archimedean place

https://alainconnes.org/wp-content/uploads/Selecta.pdf?utm_source=chatgpt.com

[3] [4] [13] [17] [18] [19] [20] [39] [41] [42] Weil positivity and Trace formula, the archimedean place

https://arxiv.org/abs/2006.13771?utm_source=chatgpt.com

[5] [9] [14] On the Number of Prime Numbers less than

https://www.claymath.org/wp-content/uploads/2023/04/Wilkins-translation.pdf?utm_source=chatgpt.com

[10] The Riemann Hypothesis: Past, Present and a Letter...

https://arxiv.org/html/2602.04022?utm_source=chatgpt.com

[11] [37] Random matrices and the Riemann zeta function

https://empslocal.ex.ac.uk/people/staff/mrwatkin/zeta/random.htm?utm_source=chatgpt.com

[15] [21] [23] [47] <https://arxiv.org/abs/0912.3183>

<https://arxiv.org/abs/0912.3183>

[22] <https://s3.cern.ch/inspire-prod-files-1/1e65b86fec7566dba4d2d2384183f67b>

<https://s3.cern.ch/inspire-prod-files-1/1e65b86fec7566dba4d2d2384183f67b>

[24] [27] [48] <https://epub.uni-regensburg.de/29572/27/1307.6055v1.pdf>

<https://epub.uni-regensburg.de/29572/27/1307.6055v1.pdf>

[25] [26] [38] An exact trace formula and zeta functions for an infinite quantum graph with a non-standard Weyl asymptotics

https://arxiv.org/abs/1104.1364?utm_source=chatgpt.com

[28] <https://www.numdam.org/item/10.5802/aif.1955.pdf>

<https://www.numdam.org/item/10.5802/aif.1955.pdf>

[29] <https://www.math.u-bordeaux.fr/~bmorin/Deninger-WA5.pdf>

<https://www.math.u-bordeaux.fr/~bmorin/Deninger-WA5.pdf>

[30] <https://www.math.purdue.edu/~branches/riemann-hilbert.pdf>

<https://www.math.purdue.edu/~branches/riemann-hilbert.pdf>

[31] [49] <https://arxiv.org/pdf/math/0202141>

<https://arxiv.org/pdf/math/0202141>

[32] <https://empslocal.ex.ac.uk/people/staff/mrwatkin/zeta/conreyRH.pdf>

<https://empslocal.ex.ac.uk/people/staff/mrwatkin/zeta/conreyRH.pdf>

[33] <https://www.jmilne.org/math/xnotes/pRH.pdf>

<https://www.jmilne.org/math/xnotes/pRH.pdf>

[34] [36] Riemann Hypothesis

https://www.claymath.org/millennium/riemann-hypothesis/?utm_source=chatgpt.com

[35] [44] The Riemann Hypothesis: Past, Present and a Letter Through Time

https://arxiv.org/abs/2602.04022?utm_source=chatgpt.com

[40] The Riemann Hypothesis: Past, Present and a Letter...

https://arxiv.org/pdf/2602.04022?utm_source=chatgpt.com

[43] Prolate operator and Riemann Zeta - Alain Connes

https://alainconnes.org/wp-content/uploads/PNAS_030322.pdf?utm_source=chatgpt.com